A Knowledge-Based Decision Support System for Context-Aware Music Recommendation

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Abstract: Traditional music recommendation systems generally rely on past listening history or aggregate popularity, ignoring the user's immediate physical and mental state. This project presents a rule-based Decision Support System (DSS) developed in SWI-Prolog that selects appropriate musical tracks based on real-time activity and biometric data.

The system merges two JSON datasets (Spotify audio features and smartwatch biometrics) into a unified Prolog knowledge base. Using backward chaining inference, it evaluates candidate tracks through physiological, cognitive, circadian, and safety constraints. In addition to selecting single optimal tracks ranked by a multi-attribute utility function, the system can synthesize gradual 3-stage relaxation sessions based on Altshuler's Iso-Principle.

1. Introduction & Motivation

Standard music recommendation engines assume user preferences are static across different situations. However, playing high-energy or fast-tempo music to someone under acute stress can worsen agitation and elevate cardiovascular strain. Conversely, during intense physical exercise like running, slow ambient music fails to sustain pace. Similarly, during cognitive study sessions, lyrical songs create phonological distractions that impair working memory.

To address these issues, this project implements a transparent, rule-based Decision Support System (DSS). By evaluating user state (heart rate, stress level, current activity, and goal) against musical properties (tempo in BPM, energy, valence, and instrumentalness), the DSS recommends tracks that actively support the user's physical and mental objectives.

2. System Architecture

The system is structured into three clear and modular components:

1. **Data Ingestion & Knowledge Base Generator (`generate_kb.py`):** Ingests raw JSON data (Spotify tracks and smartwatch biometric logs) and compiles them into normalized Prolog facts in `knowledge_base.pl` .
2. **Inference Engine & Decision Rules (`dss_rules.pl`):** Contains the backward-chaining rules and sub-rules that filter candidates, apply safety constraints, compute utility scores, and build gradual relaxation sequences.
3. **User Interface & Verification Module (`main.pl`):** Provides high-level query entry points (`recommend/1` , `recommend_session/1`), an automated demo routine (`demo/0`), and built-in unit tests (`run_tests/0`).

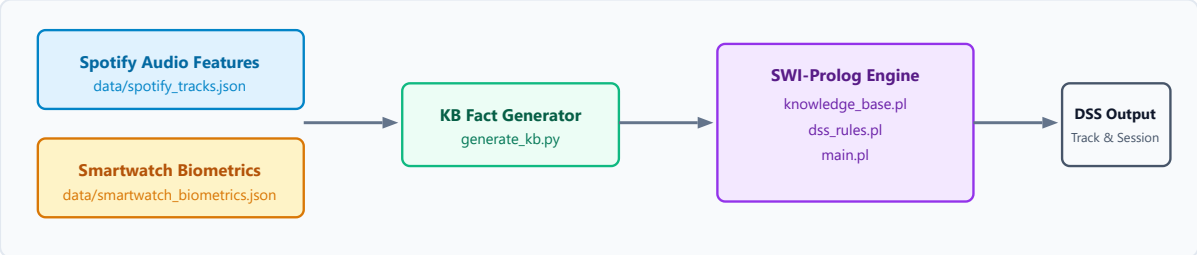


Figure 1: Knowledge-Based DSS architecture and data pipeline.

3. Knowledge Base Representation

The knowledge base represents domain entities through normalized first-order logic predicates:

3.1 Musical Entities: `track/8`

```
track(Id, Title, Artist, Genre, BPM, Valence, Energy, Instrumentalness).
```

- `BPM` : Musical tempo in beats per minute (e.g., 60.0 ambient, 140.0 techno).
- `Valence` (∈ [0.0, 1.0]): Perceived musical positiveness and cheerfulness.
- `Energy` (∈ [0.0, 1.0]): Acoustic intensity and perceived activity.
- `Instrumentalness` (∈ [0.0, 1.0]): Confidence that the track contains no vocals (≥ 0.70 means lyric-free).

3.2 User Biometric Profiles: `user/7`

```
user(Id, Name, Activity, HeartRate, StressLevel, TargetGoal, PreferredGenres).
```

- `Activity` : Current user context (`high_stress_work` , `cardio_running` , `deep_focus_study` , `sleep_preparation` , `active_commute`).
- `HeartRate` : Current pulse reading in BPM.
- `StressLevel` : Current estimated stress index on a 0–100 scale.
- `TargetGoal` : Intended objective (`stress_downregulation` , `cardio_peak_performance` , `deep_focus_flow` , `sleep_induction` , etc.).
- `PreferredGenres` : List of musical genres preferred by the user (e.g., `[classical, ambient, lofi]`).

4. Inference Engine & Decision Rules

The core decision logic in `dss_rules.pl` uses backward chaining to determine if a track is suitable for recommendation via the primary rule:

```
recommendable_track(TrackId, UserId) :-
    physiologically_compatible(TrackId, UserId),
    cognitively_compatible(TrackId, UserId),
    safety_and_contraindications(TrackId, UserId),
    circadian_coherent(TrackId, UserId).
```

4.1 The Four Constraint Pillars

1. **Physiological Compatibility (`physiologically_compatible/2`):** Matches track tempo and energy to the user's physiological requirements:
   - *Acute stress state* (elevated heart rate ≥ 100 or stress ≥ 65): requires a calming track with `BPM ≤ 75`, `Energy ≤ 0.35`, and `Instrumentalness ≥ 0.70`.
   - *Cardio running*: enforces high tempo and energetic activation (`BPM ≥ 120`, `Energy ≥ 0.70`) to match athletic cadence.
   - *Sleep preparation*: requires quiet somnolent tracks (`BPM ≤ 64`, `Energy ≤ 0.20`, `Instrumentalness ≥ 0.85`).
   - *General routine*: permits moderate tempo and activation (`70 ≤ BPM ≤ 115`, `Energy ≤ 0.65`).
2. **Cognitive Compatibility (`cognitively_compatible/2`):** When the user is engaged in study or deep focus (`deep_focus_study`), vocals create phonological interference with working memory. The rule enforces `Instrumentalness ≥ 0.70`, `70 ≤ BPM ≤ 95`, and `Energy ≤ 0.55`.
3. **Safety & Contraindications (`safety_and_contraindications/2`):** Applies negative constraints via Prolog's negation as failure (`\+`):
   - *Tachycardia constraint:* if `HeartRate > 110` and `Stress > 75`, tracks with `BPM > 100` or `Energy > 0.60` are strictly forbidden.
   - *Overstimulation constraint:* during high-stress work with `Stress > 80`, tracks with `Energy > 0.50` are barred.
4. **Circadian Coherence (`circadian_coherent/2`):** Ensures tracks for sleep preparation do not exceed `68 BPM` and `0.30 Energy`.

4.2 Progressive Relaxation: Altshuler's Iso-Principle

Abruptly presenting a very slow track (e.g., 60 BPM) to an individual with severe stress and tachycardia (114 BPM) can cause cognitive rejection. In music therapy, Ira Altshuler's Iso-Principle states that acoustic pacing must initially match the patient's current state before progressively transitioning toward the target state.

The predicate `iso_principle_session/2` dynamically synthesizes a 3-track gradual descent:

$$\text{Stage 1 [Entry]: } |\text{BPM - HeartRate}| \leq 18 \longrightarrow \text{Stage 2 [Bridge]: } 80 \leq \text{BPM} \leq 95, \text{Energy} \leq 0.50 \longrightarrow \text{Stage 3 [Target]: BPM} \leq 68, \text{Energy} \leq 0.25$$

A strict monotonicity constraint (`BPM1 ≥ BPM2 ≥ BPM3`) ensures a smooth and steady deceleration of tempo.

5. Multi-Attribute Utility Scoring

When multiple tracks satisfy all constraints, the system ranks them using `calculate_utility/3`:

$$\text{FinalScore} = \min(100.0,\ 0.50 \cdot \text{S}_{\text{frequency}} + \text{Bonus}_{\text{genre}} + \text{S}_{\text{affective}})$$

- **Target Frequency Score ($S_{frequency}$, max 100 × 0.50 = 50 pts):** Proximity of track BPM to the activity's ideal target BPM:
  `DeltaBPM = abs(BPM - TargetBPM)`, $S_{frequency}$ `= max(20.0, 100.0 - 1.5 · DeltaBPM)`.
- **Genre Preference Bonus ($Bonus_{genre}$ = 25 pts):** Awarded if the track's genre is in the user's preferred genres list.
- **Affective & Instrumentalness Score ($S_{affective}$, max 25 pts):** Weighted combination of emotional valence and instrumental purity:
  $S_{affective}$ `= (Valence · 15.0) + (Instrumentalness · 10.0)`.

The predicate `best_track/3` uses SWI-Prolog's built-in `findall/3` to evaluate all candidates and `sort/4` to select the highest-scoring track.

6. Logical Verification & Test Results

The system includes native Prolog unit tests in `main.pl`, executed via `run_tests/0`. All tests verify formal logical constraints and negative barriers:

#	Test Description	Test Condition & Input	Expected Verification	Status
1	Stress Downregulation	Sarah Miller (Stress 84, HR 114, `high_stress_work`)	Selected track has `BPM ≤ 75` and `Energy ≤ 0.35`	PASS
2	Cardio Running Cadence	Alex Chen (HR 146, `cardio_running`)	Selected track has `BPM ≥ 120` and `Energy ≥ 0.70`	PASS
3	Deep Focus Instrumentalness	Marco Rossi (`deep_focus_study`)	Selected track has `Instrumentalness ≥ 0.70` (no vocals)	PASS
4	Iso-Principle Monotonicity	Sarah Miller (Relaxation session)	Tempo decreases monotonically: `BPM1 ≥ BPM2 ≥ BPM3`	PASS
5	Safety Barrier (Negative Test)	Sarah Miller with High-Energy Rock Track	Rejected via `\+ safety_and_contraindications/2`	PASS
6	Cognitive Barrier (Negative Test)	Marco Rossi with Vocal Pop Track	Rejected via `\+ cognitively_compatible/2` during study	PASS

7. Real Execution Case Walkthroughs

The following traces represent the exact outputs produced by SWI-Prolog queries in `main.pl`:

Case 1: Acute Stress & Work Fatigue (Sarah Miller)

Profile: HR = 114 BPM, Stress = 84/100, Activity = `high_stress_work`, Goal = `stress_downregulation`.
System Decision: Selects "*Clair de Lune (Debussy)*" by Sinfonia Chambers (Score: 90.0/100, classical, 64 BPM, Energy: 0.18, Instrumentalness: 0.94).
Rationale: High stress detected. Selected a slow tempo (64 BPM) and instrumental track (0.94) to help reduce stress.
Iso-Principle 3-Stage Session:

```
1. [Entry]   Sunlit Coastal Drive         (98 BPM, Energy: 0.55)
2. [Bridge]  Analog Cassette Memory       (80 BPM, Energy: 0.38)
3. [Target]  Weightless Ambient Drift     (60 BPM, Energy: 0.12)
Trajectory: 98 BPM -> 80 BPM -> 60 BPM
```

Case 2: Cardio Running & Athletic Cadence (Alex Chen)

Profile: HR = 146 BPM, Stress = 45/100, Activity = `cardio_running`, Goal = `cardio_peak_performance`.
System Decision: Selects "*Subterranean Stamina Loop*" by Klangwerk (Score: 89.5/100, techno, 140 BPM, Energy: 0.92, Instrumentalness: 0.90).
Rationale: Cardio workout in progress. Selected a fast tempo track (140 BPM) to match the exercise pace.

Case 3: Cognitive Deep Study (Marco Rossi)

Profile: HR = 74 BPM, Stress = 32/100, Activity = `deep_focus_study`, Goal = `deep_focus_flow`.
System Decision: Selects "*Analog Cassette Memory*" by BeatFoundry (Score: 92.3/100, lofi, 80 BPM, Energy: 0.38, Instrumentalness: 0.86).
Rationale: Study/focus session. Selected an instrumental track (0.86) to avoid vocal distractions.

8. Conclusion

This project demonstrates the effectiveness of symbolic knowledge representation and rule-based backward chaining for context-aware Decision Support Systems. By combining declarative rules with first-order logic inference in SWI-Prolog, the system ensures deterministic recommendation decisions, provides clear explanations for each selection, and guarantees that physiological safety and cognitive constraints are strictly respected.